

Data Science Road Map

Module 1: Foundations of AI, ML, and Generative AI Topics

Understanding Supervised Machine Learning, Unsupervised Machine Learning, and Reinforcement Learning

Data Manipulation for Machine Learning

Exploratory Data Analysis for Machine Learning

Essential Mathematics for Machine Learning and AI Definition of Generative AI

Generative AI Applications

Definition of LLMs

Module 2: Deep dive into Machine Learning Algorithms Topics

Regression Supervised Machine Learning

Linear Regression

Decision Tree Regressor

Random Forest Regressor

Overfitting, Underfitting, and Regularization

Classification Supervised Machine Learning

Logistic Regression

Decision Tree Classifier

Random Forest Classifier

Unsupervised Machine Learning

K-Means Clustering

Hierarchical Clustering

Dimensionality Reduction

Anomaly detection

MLOps for Machine Learning model deployment—Not added resources

Resources:

https://www.youtube.com/watch?v=bTN-6VPe8c0&list=PLZoTAEIcRMXVPzj1D0i_6ajJ6gyD22b3jh

ML Supervised Learning Basics

<https://www.coursera.org/learn/machine-learning?specialization=machine-learning-introduction>

<https://www.coursera.org/learn/advanced-learning-algorithms?specialization=machine-learning-introduction>

ML Unsupervised Learning Basics

<https://www.coursera.org/learn/unsupervised-learning-recommenders-reinforcement-learning?specialization=machine-learning-introduction>

Mathematics for ML (Optional but needed. At least see once if it is difficult also.)

<https://www.coursera.org/learn/machine-learning-linear-algebra?specialization=mathematics-for-machine-learning-and-data-science>

<https://www.coursera.org/learn/machine-learning-calculus?specialization=mathematics-for-machine-learning-and-data-science>

<https://www.coursera.org/learn/machine-learning-probability-and-statistics?specialization=mathematics-for-machine-learning-and-data-science>

Module 3: Understanding Deep Learning

Topics Exploring Deep Learning Understanding Neural Networks with TensorFlow

Deep dive into Neural Networks with TensorFlow

Artificial Neural Networks (ANN)

Convolutional Neural Networks (CNN)

Object detection using the YOLO framework

Transfer Learning in CNN

Recurrent Neural Networks (RNN) and LSTMs

Module 4: Exploring NLP Topics

Architectures of RNN - One to One, Many to Many, Many to One, etc

Understanding Text for NLP

Tokenization, Stemming and Lemmatization, Stopwords, and Keywords in NLP

Text Vectorization using TF-IDF, Bag of Words, and Word 2 Vec Use cases across domains in NLP

Resources:

<https://www.coursera.org/learn/neural-networks-deep-learning?specialization=deep-learning>

<https://www.coursera.org/learn/deep-neural-network?specialization=deep-learning>

<https://www.coursera.org/learn/machine-learning-projects?specialization=deep-learning>

<https://www.coursera.org/learn/convolutional-neural-networks?specialization=deep-learning>

<https://www.coursera.org/learn/nlp-sequence-models?specialization=deep-learning>

<https://www.coursera.org/learn/classification-vector-spaces-in-nlp?specialization=natural-language-processing>

<https://www.coursera.org/learn/probabilistic-models-in-nlp?specialization=natural-language-processing>

<https://www.coursera.org/learn/convolutional-neural-networks-tensorflow>

<https://www.coursera.org/learn/natural-language-processing-tensorflow>

<https://www.coursera.org/learn/introduction-tensorflow?specialization=tensorflow-in-practice>

<https://www.coursera.org/learn/convolutional-neural-networks-tensorflow?specialization=tensorflow-in-practice>

<https://www.coursera.org/learn/natural-language-processing-tensorflow?specialization=tensorflow-in-practice>

<https://www.coursera.org/learn/tensorflow-sequences-time-series-and-prediction?specialization=tensorflow-in-practice>

<https://www.coursera.org/learn/advanced-computer-vision-with-tensorflow>

Module 5: Engaging with Generative AI Topics

How Generative AI works

Text generation, Image generation Audio generation and video generation

Using Hugging face to access models for text generation, image generation

GANs Types of GANs - ProGAN, SRGAN, CycleGAN

Auto Encoders, Variational Auto Encoders (VAEs), Diffusion Models

Module 6: Understanding LLMs

Definition of LLMs

LLM Use Cases

Prompt Tuning

Attention Mechanism

Transformer Model and Architecture

Encoder-Decoder arrangements

Train and generate text using

Encoder-decoder Architecture

BERT for Transfer learning

Leveraging multiple pretrained LLMs from Hugging Face

Fine-tuning LLM

Resources:<https://www.coursera.org/learn/introduction-to-generative-ai-for-software-development?specialization=generative-ai-for-software-development>
<https://www.coursera.org/learn/team-software-engineering-with-ai?specialization=generative-ai-for-software-development>
<https://www.coursera.org/learn/ai-powered-software-and-system-design?specialization=generative-ai-for-software-development>
<https://www.coursera.org/learn/build-basic-generative-adversarial-networks-gans?specialization=generative-adversarial-networks-gans>
<https://www.coursera.org/learn/build-better-generative-adversarial-networks-gans?specialization=generative-adversarial-networks-gans>
<https://www.coursera.org/learn/apply-generative-adversarial-networks-gans?specialization=generative-adversarial-networks-gans>
<https://www.coursera.org/learn/pytorch-fundamentals?specialization=pytorch-for-deep-learning>
<https://www.coursera.org/learn/pytorch-techniques-and-ecosystem-tools?specialization=pytorch-for-deep-learning>
<https://www.coursera.org/learn/pytorch-advanced-architectures-and-deployment?specialization=pytorch-for-deep-learning>
<https://www.coursera.org/learn/generative-deep-learning-with-tensorflow?specialization=tensorflow-advanced-techniques>
<https://www.coursera.org/learn/custom-models-layers-loss-functions-with-tensorflow?specialization=tensorflow-advanced-techniques>
<https://www.coursera.org/learn/custom-distributed-training-with-tensorflow?specialization=tensorflow-advanced-techniques>
<https://www.coursera.org/learn/advanced-computer-vision-with-tensorflow?specialization=tensorflow-advanced-techniques>
<https://www.youtube.com/watch?v=DkzbCJtFvqM&list=PLZoTAE LR MXVOTsz2jZl2Oq3ntWPoKRKwv>
<https://www.coursera.org/specializations/tensorflow-data-and-deployment> (Last Priority/Optional)

Module 7: GenAI Application Development Prompting Techniques

Topics Introduction to Prompt Engineering
Successful and Unsuccessful prompts
Types of Prompting
Introduction to Open AI, GPT, Open AI Playground
Cost & latency considerations when calling APIs (OpenAI, Azure, AWS).
Multimodal prompting for GPT 4
Image generation using Open AI DALL-E 3
Prompt evaluation Implementing Agents and Chains
Implementing zero-shot-react, conversational-react agents with LangChain
Open AI Function (Tool Integration)
Testing various LLMs with Prompt Engineering

Module 8: Synthetic Data and Datasets for LLMs

Topics Introduction to Synthetic Data
Generating Synthetic Data Synthetic Data for LLMs
Real-world Applications and Use Cases
Hands-on generating and using Synthetic Data

Module 9: AI Embeddings & Retrieval

Topics Understanding AI Embeddings
Advanced Retrieval Techniques
Hugging Face Embeddings

Vector Databases in AI
CRUD operations with Vector Databases
RAG - Retrieval Augmented Generation
RAG solutions using Open AI models and Hugging face models
Ethical Considerations in AI Embeddings
Navigating AI Hallucinations, Drift, and Bias
Embeddings in Real-world Applications
Embeddings Optimization and Fine-tuning
Embeddings Security and Privacy
LLM Ops and model deployment best practice

Resources:

<https://www.youtube.com/watch?v=CbpsDMwFG2g&list=PLZoTAE LR MX V MT W G W 9 i S 4 5 Z T c M s n t o s 6 V O>
<https://www.youtube.com/watch?v=it0l6lx3qI0&list=PLZoTAE LR MX V N b D m G Z l c g C A 3 a 8 m R Q p 5 a x b>
https://www.youtube.com/watch?v=4O1rs7mrNDo&list=PLZoTAE LR MX V O R E 4 V F 7 W Q _ f A I 0 L 1 G l j t a r
<https://www.youtube.com/watch?v=6S59Y0ckTm4&list=PLZoTAE LR MX V N 9 V b A x 5 I 2 V v l o T t Y m l A p e 3>
https://www.youtube.com/watch?v=vVGXPRjtAJE&list=PLZoTAE LR MX V P F d 7 J d v B - r n T b _ 5 V 2 6 N Y N O
<https://www.youtube.com/watch?v=1BSwYIJUxK0&list=PLZoTAE LR MX V O k 1 p R c O C a G 5 x t X x g M a l p l e>
https://www.youtube.com/watch?v=hTacGMfL8lc&list=PLZoTAE LR MX V N K t p y 0 U _ M x 9 N 2 6 w 8 n 0 h l b s
<https://www.youtube.com/watch?v=fZM3oX4xEyg&list=PLZoTAE LR MX V M 8 P f 4 U 6 7 L 4 U u D R g V 4 T N X 9 D>
https://www.youtube.com/watch?v=bFB4zqkcatU&list=PLZoTAE LR MX V O _ N f d H E B Q y l s A Y 2 y 7 8 9 p X A
https://www.youtube.com/watch?v=FH4BMLpTZuw&list=PLZoTAE LR MX V N R 8 W e q l F J Z _ 2 4 6 u N Q o 4 U d e

Note:(Need to Understand from USHA for LLM Integration)

Module 13: Container Orchestration Topics

Understanding LLM Deployment Architectures
Containerizing LLM Inference Services (e.g., using FastAPI + Docker)
Managing GPU Workloads in Kubernetes
Scaling LLM APIs with Kubernetes and Istio
Optimizing Latency and Throughput for LLM Containers
Secure Access and Rate Limiting for AI APIs
CI/CD for LLM-Powered Microservices
Monitoring and Logging for LLM Containers
Model Versioning and Rollbacks
Cost Optimization Strategies for LLM Inference in Production

Module 15: Understanding Cloud in context of Gen AI and LLMs

Topics EC2 Deep Dive and AMIs, EBS vs S3 vs EFS – Storage Solutions
Load Balancing and Auto Scaling Basics
Intro to Serverless: AWS Lambda
Using AWS Bedrock for GenAI (including foundation models)
Deploying Open-Source LLMs on EC2/EKS
Fine-Tuning and Inference
Pipelines on Cloud
Cost & Performance Considerations for LLM Workloads
SageMaker Pipelines for Model
Training and Inference
Model Versioning, A/B Testing, and Rollbacks

Security and Compliance for GenAI in Production

Deployment Resources (1.Docker+Kubernetes,2.AWS EC2,MLOps(very Important)

<https://www.coursera.org/learn/ckad-container-and-container-orchestration-fundamentals>

<https://www.coursera.org/learn/virtualization-docker-kubernetes-data-engineering>

<https://www.coursera.org/learn/docker-certified-associate-getting-started-with-docker>

<https://www.coursera.org/learn/langchain-course-for-llm-application-development>

<https://www.coursera.org/learn/open-source-llmops-solutions>

<https://www.coursera.org/learn/genai-llm-aws>

<https://www.coursera.org/learn/llmops-azure>

<https://www.coursera.org/learn/intro-gen-ai>

<https://www.coursera.org/learn/operationalizing-ml-models-mlops-for-scalable-ai>

<https://www.coursera.org/learn/mlops-mlflow-huggingface-duke>

<https://www.coursera.org/learn/mlops-aws-azure-duke>

<https://www.coursera.org/learn/certified-kubernetes-application-developer-advanced-kubernetes>

<https://www.coursera.org/learn/docker-for-the-absolute-beginner>

<https://www.coursera.org/learn/certified-kubernetes-application-developer-kubernetes-fundamentals>

<https://www.coursera.org/learn/docker-certified-associate-advanced-docker>

<https://www.coursera.org/learn/docker-certified-associate-docker-fundamentals>

<https://www.coursera.org/learn/aws-amazon-ec2-basics>

<https://www.coursera.org/learn/generative-ai-prompt-engineering-for-everyone>

<https://www.coursera.org/learn/building-gen-ai-powered-applications>

<https://www.coursera.org/learn/cloud-machine-learning-engineering-mlops-duke>

<https://www.coursera.org/learn/devops-dataops-mlops-duke>